Problem 6.19

On January 1, 2000, you invested \$1,000. Your investment grows to \$1,400 by December 31, 2007. What was the compound interest rate at which you invested?

From January 1, 2000 to December 31, 2007 is 8 years, so under compound interest,

$$1400 = 1000(1+i)^8.$$

Dividing by 1000,

$$1.4 = (1+i)^8.$$

Taking the eighth root,

$$1 + i = (1.4)^{1/8}.$$

Therefore,

$$i = (1.4)^{1/8} - 1.$$

Hence,

$$\mathrm{i} = (1.4)^{1/8} - 1 \approx 0.0429.$$

So the compound annual interest rate is approximately

$$\boxed{4.29\%}.$$